No additional strata, no loop! - A re-analysis of Malayalam compound morphophonology

Abstract: In this article, we present an analysis of the morphophonological properties of Malayalam compounds, with a particular focus on stress. Our analysis maintains the original claim of strict level ordering made in Lexical Phonology and Morphology (LPM) (Kiparsky 1982), which ensures a restrictive interleaving of morphology and phonology. The morphophonological behaviour of Malayalam compounds is first presented and analysed by Mohanan (1986) in LPM. He claims that different compound types are associated with distinct strata due to systematic phonological differences between compounds. However, compound embedding indicates that these strata must interact bidirectionally, which necessitates the introduction of a stratal loop. Introducing the loop means giving up strict ordering of strata, which results in a less restrictive architecture of grammar. Contrary to Mohanan (1986), we show that the loop is an unnecessary tool once the assumption is made that all compounding takes place in a single stratum. We explain the phonological differences in compounds by a difference in prosodic structure, thereby replacing the stratal distinction. Further, our analysis extends naturally to two cases of opaque phonology. Mohanan (1996) claims that both cases provide irrefutable evidence for the stratal distinction of compounds. However, we show that both opaque interactions follow from the proposed prosodic difference in compound types and strictly ordered strata.

Keywords: level ordering, opacity, compounds, stress, Malayalam

1 Introduction

Traditional Lexical Phonology and Morphology (LPM) (Kiparsky 1982, Mohanan 1986) and its modern counterpart Stratal Optimality Theory (OT) (Kiparsky 2000, Bermúdez-Otero 2018) assume an architecture of grammar where morphological and phonological rules[1] are distributed across strata. These strata are assumed to be strictly ordered (Kiparsky 1982, Mohanan 1986), which implies that if stratum 2 is ordered after stratum 1, then morphological and phonological rules of stratum 2 cannot precede those of stratum 1. Mohanan (1986) visualises the assumption of strict ordering with a metaphor that treats the lexicon as a word-making factory. In this factory, a word is built via a conveyor belt which leads through multiple rooms, the strata. In each room, distinct morphological (affixation, compounding etc.) and phonological rules apply to the word. Crucially, the word can travel on the conveyor belt from room 1 to room 2, but cannot traverse backwards from room 2 to room 1; the conveyor belt goes only in one direction. However, Mohanan shows that to account for the morphophonology of compounds in Malayalam, and in particular stress in compounds, the one-way directionality of the "conveyor belt" cannot be upheld. He asserts that Malayalam requires a looping mechanism to allow the derivation to revisit an earlier stratum. In this paper, we show that strict ordering in a stratal architecture can be maintained by merging Mohanan's (1986) two compound strata into a single stratum, thereby getting rid of the loop. This counters previous claims on the invalidity of stratal ordering (Orgun 1996, Inkelas 2014) and provides a strong argument to maintain a restrictive approach to morphological and phonological interleaving. Our analysis of compound stress in Malayalam follows Sproat (1986) and Inkelas (1989) who replace Mohanan's stratal distinction with a prosodic distinction which obviates the need for a loop. Further, we expand on Sproat and Inkelas by covering the entire empirical domain of compound stress and its opaque interactions with the phonological system of Malayalam (Mohanan 1996). We maintain that the attested phonological opacity neither requires the separation of compound types into distinct strata nor the loop. Instead, opacity is a reflex of prosodic structure and strict stratal ordering.

The starting point of the present paper is the analysis presented in Mohanan (1982, 1986) of the morphology and phonology of compounds in Malayalam. Alongside Pesetsky (1979) and Kiparsky (1982), Mohanan's (1986) work is attributed to the creation and development of LPM. Mohanan provides an analysis of Malayalam's morphophonology which makes use of four morphophonologically distinct lexical strata, as well as a postlexical stratum. In addition, he assumes a loop from stratum 3 back to stratum 2. His stratal architecture is presented in (1).

(1)

stratum 1: derivational affixation
↓
stratum 2: subcompounding
loop ↓
stratum 3: co-compounding
↓
stratum 4: inflectional affixation
↓
postlexical stratum

The motivation to distinguish four lexical strata, and in particular to distinguish subcompounds (tatpuruṣas) from co-compounds (dvandva), comes from a difference in their phonology. The examples in (2) show the compounding of two nouns *box* and *grainbin*. All examples presented in this paper come from prior research by Mohanan (1982, 1986). In (2a), the two nouns form a subcompound, which entails a structure such that *grainbin* is the head of the compound and *box* is its modifier. In (2b), the two nouns form a coordinative compound. A co-compound has a structure where both nouns are treated as heads, which leads to a symmetric, coordinative interpretation (see Wälchli 2005, Ralli 2019 for an overview on coordinative compounds). The two types of compounds differ not only in their morphological structure but also in their surface phonology. Firstly, the two compound types surface with different stress domains. Subcompounds surface as one stress domain, indicated by the bracketing, while co-compounds surface as two stress domains. Secondly, subcompounds undergo lengthening processes that co-compounds do not. In (2a), a derived geminate consonant, boxed for convenience, surfaces at the subcompound boundary. In (2b) no geminate surfaces in the co-compound.

(2) a. [(ˈpèʈʈipp at̪ˌt̪áːjáŋ)ŋalə]
petti- pat̪t̪aːjam -kaɭ
box- grainbin -PL
'grainbins used as boxes' [subcompound] (p.90)[2]

b. [(ˈpèʈʈí)($\boxed{\text{p}}$ at̪ˈt̪à:jáŋ)ŋalə]

peʈʈi- pat̪t̪a:jam -kaɭ

box- grainbin -PL

'boxes and grainbins' [co-compound] (p.90)

Mohanan takes these phonological differences to mean that sub- and co-compounds undergo distinct phonological rules, which in LPM means that the two compound types are built at distinct morphological strata. He proposes that nouns subcompound at stratum 2, where they undergo the rule of stress assignment as a compounded unit. Here, a subcompound also undergoes other rules, such as stem initial consonant gemination. Nouns that co-compound only do so at the next stratum, at stratum 3, where rules such as stem initial consonant gemination are turned off and stress assignment has already applied to the individual nouns at stratum 2, prior to stratum 3 co-compounding. This analysis is sketched below in (3) with the examples of (2).

(3) *Stratal derivation of Malayalam compound morphophonology in Mohanan (1986):*

	subcompound	co-compound
stratum 2 morphology		
subcompounding	[[peʈʈi] [pat̪t̪a:jam]]	N/A [peʈʈi] [pat̪t̪a:jam]
stratum 2 phonology		
initial consonant gemination	[[peʈʈi] [ppat̪t̪a:jam]]	N/A
stress assignment	[[ˈpèʈʈi] [ppat̪ˈt̪á:jám]]	[ˈpèʈʈí] [pat̪ˈt̪à:jám]
stratum 3 morphology		
co-compounding	N/A [ˈpèʈʈippat̪ˈt̪á:jám]	[[ˈpèʈʈí] [pat̪ˈt̪à:jám]]
final result:	[ˈpèʈʈippat̪ˈt̪á:jáŋŋalə]	[ˈpèʈʈípat̪ˈt̪à:jáŋŋalə]

However, contrary to what the phonology indicates, the morphology suggests that both compound types are built at a single morphological stratum. Evidence for this comes from compound embedding. The examples in (4) show a subcompound embedded in a co-compound in (4a) and a co-compound embedded in a subcompound in (4b). In the remainder of this paper, we refer to embedded compounds as complex compounds.

(4) a. [[ˈkàːmukíː [ˈb^{h}àːrjáːsaˌhóːd̪arí]] maːrə]

kaːmuki-	b^{h}aːrja-	sahoːd̪ari	-maːr
lady.love-	wife-	sister	-PL

‘lady love and wife’s sister’ [co-compound[subcompound]] (p.123)

b. [[ˈjàːt̪íˈmàt̪á] wid̪ˈwèːʂám]

jaːt̪i-	mat̪am-	wid̪weːʂam
caste-	religion-	hatred

‘hatred of religion and caste’ [[co-compound]subcompound] (p.123)

The possibility for bidirectional embedding as in (4) implies that the two types of compounds interact morphologically. Therefore, word formation of the two compounds should apply at the same morphological stratum.

Mohanan himself recognises that the phonological and morphological evidence is in conflict with respect to the stratal association of compounds. In order to accurately reflect that the phonologies between sub- and co-compounds differ while retaining the possibility for bidirectional embedding, he introduces a stratal loop, repeated as an excerpt of (1) in (5) below.

(5) loop

stratum 2: subcompounding
↓
stratum 3: co-compounding

The stratal architecture in (5) derives that subcompounds are phonologically distinct from co-compounds by associating them to different strata. Simultaneously, (5) allows both types of embedding by virtue of the loop. However, the loop is a direct contradiction to the strict ordering of strata, which is argued to be an integral part of LPM (Kiparsky 1982). Essentially, the loop allows morphological and phonological rules of a later stratum to apply before those of an earlier stratum, deemed impossible under strict ordering. Yet, Malayalam seems to require the loop for the particularly troublesome case in (4b) where a co-compound is embedded in a subcompound. In this construction, the loop allows the grammar to first build a co-compound at stratum 3 and then loop back to stratum 2 where a subcompound can embed the co-compound.

Immediately after its introduction, the loop received criticism for weakening the theory of LPM (Sproat 1985) and a handful of re-analyses attempt to tackle the complex morphophonology of Malayalam compounds, see Sproat (1986) and Inkelas (1989). These re-analyses can explain the part of Malayalam morphophonology in (2) by doing away with the loop and, instead, introducing a prosodic difference to the phonology of compounds. However, Mohanan's (1996) reply to these re-analyses makes it clear that they fail to derive intransparent stress domains in complex compounds, as well as two cases of phonological opacity in co-compounds which do not occur in subcompounds. Thus, Mohanan (1996) argues that the loop remains essential for Malayalam compound morphophonology and, therefore, must be an integral part of the theory of LPM. More recently, the complexities of Malayalam compound morphophonology have been taken as evidence to completely abandon any kind of level ordering and instead resort to construct-specific co-phonologies, see in particular Inkelas (2014) who discusses the Malayalam data. In such a co-phonology theory (Orgun 1996, Inkelas & Orgun 1998), each morphological operation introduces a phonological cycle with its own phonological grammar, its own co-phonology, that can potentially differ from the general phonology of the language. There is no claim on an upper limit of co-phonologies and any kind of strict level ordering is completely abandoned. In the rest of this article, we establish that strict level ordering can be maintained and therefore contend that Malayalam compound morphophonology supports the architecture of stratal analyses and obviates a retreat to unconstrained co-phonologies. We do this by replying to Mohanan (1996) and expanding on Sproat (1985) and Inkelas (1989) who argue that there is no need for a loop once the compounds are no longer distinguished stratally. We argue that the distinct phonological behaviour, as presented in (2), can be explained by a difference in prosodic structure, which follows from the different morphological structures of the compounds. This assumption allows for a collapse of Mohanan's (1986) stratum 2 and 3 into a single stratum where bidirectional compound embedding is no longer an issue. Further, we tackle the problematic phonology of complex compounds, as well as phonological opacity in co-compounds. We show that despite Mohanan's (1996) claim that these cases provide irrefutable evidence for the loop, both cases can be derived without it, as long as prosodic structure is taken into account and crucially strict ordering of strata is maintained. We couch the analysis into a stratal architecture following Bermúdez-Otero (2018) that is limited to three strata, the stem-, word- and a postlexical stratum pace Kiparsky (1982), Halle & Mohanan (1985), and Mohanan (1986). The proposed strata and their strict order is shown in (6).

(6) *New stratal architecture of Malayalam:*

stem-stratum: derivational affixation, compounding
↓
word-stratum: inflectional affixation
↓
postlexical stratum

The rest of the article is structured as follows. Section 2 introduces the stress properties of Malayalam. Section 2.1 deals with deriving the stress domains in simplex compounds, and section 2.2 expands the analysis to complex compounds. In both sections, we argue that the prosodic difference leads to all phonological differences observed in both compound types and is derived from the compounds' distinct morphological structure. Section 3 introduces two instantiations of phonological opacity that arise only in co-compounds. Again, we show that this phonological effect is a reflex of the different prosodic structures of the compound types, as well as an instantiation of a cycle realised by the minimized and strictly ordered stratal architecture in (6).

2 Stress in Malayalam

Mohanan (1986) shows that Malayalam[3] has a stress system with unconventional correlates. Primary stress correlates with a low pitch. Primary stress falls on the first syllable, see (7a), except if the first syllable is short and the second syllable is long. In that case, shown in (7b), primary stress falls on the second syllable.[4] Secondary stress correlates with an H pitch and is realised on any long vowel that follows the primary stressed syllable, see (7c). Finally, a H pitch is realised on the final syllable of each word in (7). We treat this final H pitch as a H boundary tone.

(7)	a.	[ˈpàraṭ̪í]	b.	[paˈràːṭ̪í]	c.	[ˈpàːˌráːjaɳám]
		'searched'		'complaint' (p.112)		'reading' (p.113)

Although not noted by Mohanan (1986), all examples he presents indicate that primary stress does not fall onto the second syllable containing a long vowel, if that syllable is word-final. Evidence for this comes from the bisyllabic word *what* in the sentence in (8) below.[5] Its final long vowel does not surface with primary stress despite canonically being in the correct position, the second syllable of the word. Under an Optimality Theoretic approach, the impossibility of a final stressed syllable can be captured by the constraint NON-FINALITY(FT´) (Ito & Mester 2016),[6] which will become relevant to capture one of the opaque interactions in section 3.

(8) ˈmàːliní ˈmàːlá $\boxed{\text{ˈèn̪d̪éː}}$ ˈcèjt̪at̪ə́
Malini necklace what did
What did Malini do with the necklace?(pp.113-114)

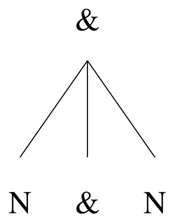
2.1 Stress domains in simplex compounds

As mentioned in section 1, sub- and co-compounds supply different domains for stress, see (7) repeated in (9) below. A subcompound in (9a) is treated as a single domain for stress whereas, each head in a co-compound in (9b) is treated as a single domain for stress. The domains are revealed by the single primary stress in the subcompound versus a single primary stress on the respective heads of the co-compound. In both cases, the inflectional affixes are not part of the stress domain; they do not surface with the H boundary tone. Instead, the H boundary tone lands on the final syllable of the compound. It follows that the domain for stress in subcompounds is the entire compound, whereas, in co-compounds, the domains for stress are the individual nouns, excluding inflectional affixes in both compound structures.

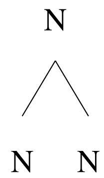
(9)	a.	[(ˈpèʈʈippat̪ˌt̪áːjáŋ)ŋalə]	b. [(ˈpèʈʈí)(pat̪ˈt̪àːjáŋ)ŋalə]
		peʈʈi- pat̪t̪aːjam -kaɭ	peʈʈi- pat̪t̪aːjam -kaɭ
		box- grainbin -PL	box- grainbin -PL
		'grainbins used as boxes'	'boxes and grainbins' (p.90)

We argue that what differs between sub- and co-compounds is not their association to distinct morphological strata as Mohanan (1986) suggests. Instead, what differs is the morphological structure of sub- and co-compounds, which maps onto distinct prosodic structures that impose distinct stress domains. In detail, we propose that co-compounds have the morphological structure in (10a), a symmetric nominal compound headed by an abstract coordination head, which we label '&' in accordance with the practise of current minimalist literature (Zoerner 1995, Zhang 2010, Weisser 2015). For independent arguments to assume a functional element as the head of compounds see Di Sciullo (2005, 2011) and to assume an abstract coordination head in co-compounds see Nóbrega (2020). We assume that the morphosyntactic features of nouns in a co-compound structure transfer to the &-head and percolate, see Murphy & Puškar (2018) for a concrete implementation of such feature inheritance as syntactic agreement. This allows the coordination constituent to be treated like any nominal head in terms of inflectional affixation and further compounding. The structure of co-compounds in (10a) leads to the prosodic parse in (11a), where both nouns are parsed as their own separate prosodic words. Under the assumption that the minimal prosodic word is the domain for stress, this parsing ensures that each noun in a co-compound receives its own primary stress in the form of an L tone and its own boundary H tone on the final syllable. We assume that the tonal correlates of stress are assigned to the prominent positions of the minimal prosodic word in stem-level phonology. Subcompounds have the morphological structure in (10b), a balanced nominal compound structure. This structure parses into a single prosodic word encompassing both nominal elements in the compound structure in (11b). Parsing a single prosodic word ensures that in subcompounds, there is only a single primary stress and a single final boundary H tone.

(10) a. *Morphological structure of a co-compound:*



b. *Morphological structure of a subcompound:*



(11) a. *Parsing of a co-compound:*

$(_{\omega}$ N) $(_{\omega}$ N)

b. *Parsing of a subcompound:*

$(_{\omega}$ N N)

The exclusion of inflectional affixes from the domain of stress is accomplished if all inflectional affixation is assigned to the morphology of the word-stratum. In word-stratum phonology, the prosodic domains established for compounds at stem-level are retained via faithfulness constraints and therefore the positions of stress and of the boundary tone do not change. In section 3.2, we show that inflectional affixes do not integrate into minimal prosodic words and instead are parsed into maximal prosodic words in a recursive prosodic word structure. [7]

To arrive at the parses in (11) from the morphological structures in (10), we adopt MATCH constraints (Selkirk 2011) that match morphosyntactic structure to prosodic structure. Further, we base the computation in Harmonic Grammar (HG, Legendre, Miyata & Smolensky 1990, Potts et al. 2010), which differs from Optimality Theory (OT, Prince & Smolensky 1993) by having weighted and not ranked constraints. If an output violates a constraint in HG, the weight of the constraint is added to a harmony score of that output candidate. The harmony scores of each candidate are compared and the candidate with a harmony score closest to zero is deemed most harmonic. The choice of HG as opposed to OT is motivated by a ganging-up effect in the prosodic parsing of complex compounds in section 2.2. The relevant constraints that are needed to arrive at the parses in (11) are presented in (12)-(15). The first constraint in (12) constitutes a typical MATCH constraint. We diverge from Selkirk (2011) in directly referencing the nominal category, as opposed to referencing an abstract word X that should coincide with the edges of a prosodic word ω. The constraint MATCH(N,ω) in (12) enforces parsing of morphological constituents of type N into prosodic words. For example, an input such as the subcompound structure in (10b), $[_N\ [_N$ noun] $[_N$ noun]], perfectly satisfies the constraint MATCH(N,ω) if all N constituents are parsed into matching prosodic words, resulting in the prosodic structure $(_\omega\ (_\omega$ noun) $(_\omega$ noun)). The constraint will be penalised for any N constituent that is not parsed into a prosodic word. The structure $(_\omega$ noun $(_\omega$ noun)) for example, does not parse the first embedded N constituent into a prosodic word that matches the edges of that N constituent and therefore incurs a single violation of MATCH(N,ω).

(12) $\boxed{\text{MATCH(N,}\omega\text{)}}$: Assign a violation for every constituent type N whose left and right edges in the input morphological representation do not correspond to the left and right edges of a constituent of type ω in the output phonological representation. weight = 4

The second relevant constraint is the NONRECURSIVITY constraint in (13). It is penalised by parsing prosodic words inside of prosodic words thereby creating a recursive prosodic word. As an example, take again our subcompound input structure in (10b), $[_{\mathrm{N}}\ [_{\mathrm{N}}$ noun] $[_{\mathrm{N}}$ noun]]. Parsing this to completely satisfy MATCH(N,ω) creates a recursive prosodic word, cf. ($_{\omega}$ ($_{\omega}$ noun) ($_{\omega}$ noun)). This prosodic structure violates the constraint NonRec. An example for a prosodic structure given the input $[_{\mathrm{N}}\ [_{\mathrm{N}}$ noun] $[_{\mathrm{N}}$ noun]] that satisfies NonRec is ($_{\omega}$ noun noun), where there is no recursive structure.

(13) $\boxed{\text{NonRecursivity (NonRec)}}$: Assign a violation for every prosodic constituent X that dominates a prosodic constituent Y, such that X=Y (cf. Selkirk 1995).

weight = 9

The constraint MATCH(ω,N) in (14) enforces parsing of prosodic words only where there are constituents of type N. For our case, this constraint is penalised when prosodic words are parsed to match functional constituents, like the &-head. Take as an input the co-compound structure in (10a), $[_{\&}\ [_{\mathrm{N}}$ noun] $[_{\&}$] $[_{\mathrm{N}}$ noun]]. An output prosodic structure that parses a prosodic word that does not match the edges of any N constituents, like the structure ($_{\omega}$ noun noun) which parses the coordination constituent into a prosodic word, violates the constraint MATCH(ω,N) once. The constraint is satisfied if all output prosodic words match N constituents, like in the an output ($_{\omega}$ noun) ($_{\omega}$ noun) from an input co-compound structure.

(14) $\boxed{\text{MATCH}(\omega\text{,N})}$: Assign a violation for every constituent type ω whose left and right edges in the output phonological representation do not correspond to the left and right edges of a constituent of type N in the input morphological representation.

weight = 10

Finally, MATCH($\mathrm{N}^{\max}$,ω) in (15) is penalised if a specific type of constituent, a maximal N, is not matched with a prosodic word ω. A maximal N is an N constituent that is not immediately dominated by another N constituent. Let us exemplify the relevance of this constraint for the case of an input subcompound structure in (10b), $[_{\mathrm{N}}\ [_{\mathrm{N}}$ noun] $[_{\mathrm{N}}$ noun]]. Here, there is a maximal N constituent, an N constituent that is not immediately dominated by another N constituent. The constraint MATCH(N^{max},ω) is penalised if this constituent is not parsed into a prosodic word, like in an output structure ($_\omega$ noun) ($_\omega$ noun). Once the maximal N constituent is parsed into a prosodic word the constraint is satisfied, such as in a prosodic structure ($_\omega$ noun noun).

(15) $\boxed{\text{MATCH}(N^{max},\omega)}$: Assign a violation for every constituent of type N^{max} whose left and right edges in the input morphological representation do not correspond to the left and right edges of a constituent of type ω in the output phonological representation, such that N^{max} is an N that is not immediately dominated by another N (cf. Selkirk & Shen 1990). weight = 50

The computations of the prosodic parse of both compound structures are presented in tableaux (16) and (17). We begin with the computation of subcompounds in (16), the input morphological structure is that in (10b). Three output candidates are considered, the first in (16a) is the most faithful parse, where all N constituents are parsed into prosodic words, such that their edges match the edges of the prosodic words which fully satisfies both MATCH(N^{max},ω) and MATCH(N,ω). This leads to a recursive prosodic word which violates the constraint NONREC and produces harmony score of -9. The more harmonic output candidate is (16b), which parses the entire structure into a single prosodic word and thereby satisfies MATCH(N^{max},ω). This output incurs two violations of MATCH(N,ω) due to the edges of two N constituents not matching the edges of any prosodic words ω in the output. Even with these two violations, candidate (16b) is most harmonic with its harmony score of -8 because of the low weight of MATCH(N,ω). Compare this to output (16c), which avoids one MATCH(N,ω) violation by parsing two N constituents into two prosodic words. Doing this incurs a fatal violation of high-weighted MATCH(N^{max},ω) and a harmony score of -54.

(16) *Prosodic parsing of a subcompound:*

	$[_N$ N N]	MATCH(N^{max},ω) 50	NONREC 9	MATCH(N,ω) 4	$\mathcal{H}$
	a. $(_\omega\ (_\omega$ N) $(_\omega$ N))		1		-9
☞	b. $(_\omega$ N N)			2	-8
	c. $(_\omega$ N) $(_\omega$ N)	1		1	-54

In co-compounds, the computation favours a different parse given the difference in morphological structure. In subcompounds, a single prosodic word is most harmonic. In co-compounds, parsing two prosodic words is most harmonic, shown by candidate (17c) in the tableau below. Crucially, (17c) does not violate any of the constraints presented in (12)-(15). It satisfies all MATCH constraints by parsing the two available Ns into individual prosodic words. It also satisfies NONREC by not exhibiting any recursive prosodic structure. In comparison to this, the outputs in (17a) and (17b) both violate MATCH(ω,N) by virtue of parsing a prosodic word which does not match the edges of any N node, this becomes fatal in both cases.[8]

(17) *Prosodic parsing of a co-compound:*

	$[_\&$ N & N]	MATCH(ω,N) 10	NONREC 9	MATCH(N,ω) 4	$\mathcal{H}$
	a. $(_\omega\ (_\omega$ N) $(_\omega$ N))	1	1		-19
	b. $(_\omega$ N N)	1		2	-18
☞	c. $(_\omega$ N) $(_\omega$ N)				0

The Harmonic Grammar (HG) computations in (16) and (17) show that the different stress domains follow from a difference in morphological structure and MATCH constraints. Thus, the prosodic domains of simplex compounds can be computed in a parallel HG. This demonstrates that sub- and co-compounds need not be distinguished stratally. Without a stratal distinction between sub- and co-compounds the loop loses its function.

In addition to explaining the different stress domains of Malayalam compounds, the prosodic parse can also explain the application and non-application of consonant lengthening at the compound boundary that is briefly mentioned in section 1. While we cannot do full justice to the details of the lengthening effects at compound boundaries, it will suffice to mention that the non-application of consonant lengthening can be treated as a reflex of the prosodic structure in co-compounds. We assume that stem-initial gemination is blocked due to the co-compound's internal prosodic word boundary. This boundary is not present in the prosodic structure of subcompounds and therefore, a consonant geminate freely surfaces in subcompounds. For a detailed analysis of the lengthening effects see Author & Author (2025). In the next section, we show that the prosodic domains in complex compounds are computed in parallel by implementing the same constraints that derive the prosodic structure of simplex compounds. This indicates further that the loop is an unnecessary device, as it is not required to derive the stress domains of complex compounds.

2.2 Stress domains in complex compounds

Mohanan (1986) gives a handful of examples of complex compounds, where either a subcompound is embedded in a co-compound, as in example (18) or a co-compound is embedded in a subcompound in example (19). In (18), the nouns *wife* and *sister* form a subcompound that is translated as *wife's sister*. This subcompound is then co-compounded with the noun *lady love*, which results in the complex compound translated as *lady love and wife's sister*. The stress domains in this example are transparent. The subcompound *wife's sister* supplies a single stress domain, as indicated by the single primary stress on the first long syllable of *wife*. Additionally, each head in the co-compound, *lady love* and *wife's sister*, constitutes its own independent stress domain, again indicated by the single primary stress on *lady love* and on the subcompound *wife's sister*.

(18) [(ˈkàːmukíː)(ˈbʰàːrˌjáːsaˌhóːd̪arí)maːrə]

ka:muki- bʰa:rja- saho:d̪ari -ma:r

lady.love- wife- sister -PL

'lady love and wife's sister' [co-compound[subcompound]](p.123)

In the example (19) the nouns *caste* and *religion* form a co-compound. This co-compound is then subcompounded with the noun *hatred* to form the complex compound *hatred of religion and caste*. Interestingly, each noun in the complex compound surfaces with its own primary stress, which indicates that each noun is parsed into a separate prosodic word. Thus, somewhat unexpectedly, the subcompound, which embeds a co-compound, does not supply a single stress domain.

(19) [(ˈjàːt̪í)(ˈmàt̪á)(wid̪ˈwèːʂám)]

ja:t̪i-mat̪am-wid̪we:ʂam

caste-religion-hatred

'hatred of religion and caste' [[co-compound]subcompound] (p.123)

Mohanan's (1986) stratal architecture derives the stress domains of (19) with the assumption of the loop. His analysis is sketched in (20) below.

(20) *Sketch of Mohanan's (1986) analysis of (19)*

str 2 subcompounding: N/A	[ja:t̪i] [mat̪am] [wid̪we:ʂam]
str 2 phonology: stress assignment	[ˈjàːt̪í] [ˈmàt̪ám] [wid̪ˈwèːʂám]
str 3 co-compounding	[[ˈjàːt̪í] [ˈmàt̪ám]] [wid̪ˈwèːʂám]
str 3 phonology	[[ˈjàːt̪í] [ˈmàt̪ám]] [wid̪ˈwèːʂám]
loop to str 3 subcompounding	[[ˈjàːt̪íˈmàt̪á] [wid̪ˈwèːʂám]]

According to Mohanan, subcompounding and co-compounding happen at distinct strata where subcompounding precedes co-compounding. Thus in the derivation of (19), the nouns' first visit to stratum 2 cannot yet yield a subcompound, as the embedded co-compound has not yet been built. Nevertheless, all nouns undergo the phonology of stratum 2, which assigns stress. Because none of the nouns have been compounded to form a larger domain, all nouns individually receive primary and secondary stresses. Stratum 3 morphology follows, where a co-compound can be built which undergoes stratum 3 phonology.[9] After stratum 3, with the help of the loop, the grammar can revisit stratum 2. At this stage, the subcompound can be built. However, it is too late for the subcompound to be treated as a single domain for stress. What is essential to Mohanan's (1986) analysis of co-compounds embedded in subcompounds is that stratum 2 phonology (stress assignment) applies before stratum 2 morphology (subcompounding). This is only possible due to the mechanism of the loop. It ensures that, under specific morphological circumstances, stratum 2 morphology is revisited after stratum 2 and 3 phonology has already been applied, which leads to the intransparent stress domains in complex compounds in (19).

Mohanan (1986) relies on a cyclic analysis where morphology and phonology can interleave and strict ordering of strata is relaxed. We assume a parallel prosodification of all sub- and co-compounds and by extension complex compounds. The intransparent stress effects in (19) seem to be a knock-down argument against a parallel theory to Malayalam compound morphophonology. However, we exemplify that the stress domains follow from our model of prosodification owing to the computation of Harmonic Grammar. First, we present the prosodic parsing of a co-compound that is embedded in a subcompound, à la (19) with the structure $[_{N}\ [_{\&}\ N\ \&\ N\ ]\ N\ ]$. The related tableau is presented in (21). The most harmonic candidate is (21d), which is the most faithful parse of the complex compound, where each N is parsed such that its edges coincide with the edges of a prosodic word ω, yielding a recursive prosodic word. While this candidate violates NonRec, this violation is not fatal. This is a crucial departure from the derivation of simple subcompounds, where precisely this violation of NonRec ends up being fatal, see tableau (16) in the previous section. The crucial difference to simple compounds is the number of violations of MATCH(N,ω) that the alternative parse in candidate (21b) receives. (21b) parses the entire complex compound into a single prosodic word. By doing so, it incurs three violations of MATCH(N,ω) because there are three nominal elements involved in the morphological structure whose edges do not coincide with the edges of prosodic words. Compare this to the two nominal elements in the simple subcompound, leading to only two violations of MATCH(N,ω). Thus, in complex compounds, multiple violations of the low-weighted constraint MATCH(N,ω) lead to a gang effect (Pater 2009, 2016), which worsens the harmony score of candidate (21b) to -12. Candidate (21d) only violates higher-weighted NONREC once, which leads to a more harmonic score of -9. Other prosodic parses that are presented by candidates (21a) and (21c) that avoid MATCH(N,ω) violations, cause violations of higher-weighted constraints and therefore do not improve the harmony score enough to win against the most harmonic (21d).

(21) *Prosodic parsing of a co-compound embedded in a subcompound:*

$[_N [_\& N \& N] N]$	M(N^{max},ω) 50	M(ω, N) 10	NONREC 9	M(N,ω) 4	$\mathcal{H}$
a. $(_\omega (_\omega N) (_\omega N) N)$			1	1	-13
b. $(_\omega N N N)$				3	-12
c. $(_\omega N) (_\omega N) (_\omega N)$	1			1	-54
☞ d. $(_\omega (_\omega N) (_\omega N) (_\omega N))$			1		-9

For completeness sake, the tableau of the computation of subcompounds embedded in co-compounds is presented in (22). Here, the parsing works parallel to the individual parsing of sub- and co-compounds. The most harmonic output (22d) parses the embedded subcompound into a single prosodic domain, thereby the parse avoids a recursive prosodic structure while simultaneously parsing the maximal N into a prosodic word. Additionally, it parses the individual N constituents of the co-compound into separate prosodic words. Crucially, the more faithful parse in (22a) that matches all constituents of type N into prosodic words is less harmonic. Just as in simple subcompounds this parse incurs a fatal violation of NONREC. Importantly, this violation is avoided in (22d) without leading to the gang effect that was observed in tableau (21) because there are only two nouns that are part of the subcompound structure compared to the three nouns of the subcompound structure in (21).

(22) *Prosodic parsing of a subcompound embedded in a co-compound:*

$[_{\&}$ N & $[_{N}$ N N]]	M(N^{max},ω) 50	M(ω, N) 10	NonRec 9	M(N,ω) 4	$\mathcal{H}$
a. $(_{\omega}$ N) $(_{\omega}$ $(_{\omega}$ N) $(_{\omega}$ N))			1		-9
b. $(_{\omega}$ N N N)		1		2	-18
c. $(_{\omega}$ N) $(_{\omega}$ N) $(_{\omega}$ N)	1		1	2	-67
☞ d. $(_{\omega}$ N) $(_{\omega}$ N N)				2	-8

The computations of this and the previous section present a unified analysis of stress domains in Malayalam in a parallel Harmonic Grammar that makes use of prosodic structure to arrive at the relevant prosodic domains for stress. Neither the parsing of simplex compounds nor that of complex compounds necessitates a stratal distinction between the compounds of Malayalam. Without this stratal distinction, there is no need for a loop.

3 Phonological opacity in co-compounds

Mohanan (1986) presents two cases of phonologically opaque surface forms which appear in co-compounds but not in subcompounds. Our assumption is that all compound types are derived at the same stratum, the stem stratum. This assumption conflicts with the observed opaque phonology in co-compounds for two reasons. The first reason is that the phonology of subcompounds and co-compounds should not be different. Given that both compounding types undergo the same stem-level phonology, phonological effects in co-compounds should be the same as in subcompounds. The second reason is that the opaque forms in co-compounds would constitute a case of intrastratal opacity. Kiparsky (2000) and Jaker & Kiparsky (2020) claim that intrastatal opacity is impossible in a stratal OT framework.

The first issue is immediately refuted under the assumption that prosodic structure causes different phonological effects in co-compounds compared to subcompounds despite their affiliation to a single stratum. The second issue is more problematic for the proposed minimized stratal proposal. Yet, we show that our stratal architecture can account for both opaque interactions in co-compounds using the established prosodic difference in compounds and a phonological cycle instantiated by word level phonology which follows stem level phonology.

3.1 Case 1: Final vowel lengthening and stress assignment

The first case of opacity presented by Mohanan (1986) concerns the process of final vowel lengthening and its interaction with stress assignment. Final vowel lengthening occurs in all types of nominal compounds and is described as follows: At the boundary between two compounds, a final vowel of the first stem is lengthened if the second stem has Sanskrit origin.[10] (23c) exemplifies the subcompound of the nouns *bride* and *house*. In isolation, the noun *bride* surfaces with a short final vowel, see (23a). In the subcompound structure in (23c), the final vowel is lengthened. This lengthening interacts with stress placement. It attracts primary stress, boxed for convenience, because the lengthened vowel is in the second syllable of the word and the first syllable is short.

(23) a. [ˈwàd̪ʰú]
'bride'

b. [ˈgrə̀hám]
'house'

c. [($_\omega$ waˈd̪ʰ ùː grəhám)]
wad̪ʰu-grham
bride-house$_{\text{[SANSKRIT]}}$
'bride's home' [subcompound] (p.120)

To arrive at (23c) in a rule-based framework, the rule for final vowel lengthening must precede that of stress placement. This ensures that the lengthened vowel attracts stress. This interaction is one of shifting (Rasin 2022), where the application of the first rule influences the way the second rule is applied.

In co-compounds, the shifting interaction is not available. This becomes evident in example (24c) where the nouns *poet* and *poetess* are co-compounded. The final vowel of the first stem *poet* lengthens but does not surface with the expected primary stress. The lengthened vowel in (24c) surfaces with a (boundary) H tone instead.

(24) a. [ˈkàwí]
'poet'

b. [ˈkàwajit̪ɾí]
'poetess'

c. [($_\omega$ ˈkàw$\boxed{\text{íː}}$)($_\omega$ ˈkàwajit̪ɾí)kaɭə]
kawi-kawajit̪ɾi-kaɭ
poet-poetess$_{[\text{SANSKRIT}]}$-PL
'poets and poetesses' [co-compound] (p.121)

The surface form in (24c) follows from ordering the rules such that stress assignment precedes final vowel lengthening. This ordering leads to a countershifting interaction (Rasin 2022) where the second rule is expected to affect the way the first rule applies, but it ends up not doing so.

Thus, the ordering between final vowel lengthening and stress assignment is different depending on what type of compound is being dealt with. The conflict in rule ordering is resolved by Mohanan's (1986) stratal distinction between the compounds, sketched in (25). At stratum 2, where nouns are subcompounded, final vowel lengthening applies before stress assignment. At stratum 3, where nouns are co-compounded, final vowel lengthening applies again and is thus ordered after stress assignment (of stratum 2 phonology).

(25) *Rule ordering to account for opacity:*

	Morphology	Phonology
stratum 2	subcompounding	1. final vowel lengthening
		2. stress assignment
stratum 3	co-compounding	final vowel lengthening

Mohanan's (1986) analysis relies on the stratal distinction between subcompounds and co-compounds. However, in order to arrive at the surface structures in (23c) and (24c) a stratal distinction is not necessary. The difference in phonological opacity follows directly from the proposed prosodic difference of the compounds and an additional constraint, NON-FINALITY(FT´) introduced in section 2, on the placement of primary stress in prosodic words, defined in (26).

(26) NON-FINALITY(FT´): Assign a violation to every rightmost foot head that is final in a prosodic word (Ito & Mester 2016). weight = 50

This constraint receives high weight throughout the grammar of Malayalam and penalises any foot head that is rightmost in the prosodic word. Recall that the final lengthened vowel in a co-compound structure is in exactly this position, see (27a). In subcompounds, the final lengthened vowel is not in a rightmost prosodic word final position, see (27b), therefore primary stress can be realised on the lengthened vowel.

(27) a. [($_\omega$ ˈkàw$\boxed{\text{íː}}$)($_\omega$ ˈkàwajit̪ɾí)kaɭə]   b. [($_\omega$ waˈd̪ʰ$\boxed{\text{ùː}}$gɾəhám)]

The following tableau in (28) shows the computation of the prosodic structure and its interaction with stress placement and final vowel lengthening for co-compounds. We assume that the lengthening arises by the introduction of a floating mora between the two nouns[11]. Candidate (28a) integrates the floating mora, thereby creating the final lengthened vowel. However, in this candidate, primary stress does not fall on the lengthened vowel which non-fatally violates the weight-to-stress constraint WTS. Candidate (28b) realises primary stress on the long vowel, thereby incurring a fatal violation of high-weighted NON-FINALITY(FT´) constraint. Alternatively, the prosodic parse can be changed to avoid a prosodic word boundary between the two compounded nouns. This is indicated by the less harmonic candidate (28c), which violates MATCH(ω, N) by parsing a prosodic word which does not match the edges of an N constituent. A violation of NON-FINALITY(FT´) is also averted by not integrating the floating mora in candidate (28d) but doing so fatally violates *FLOAT. Thus, candidate (28a) is most harmonic despite not abiding by the weight-to-stress principle, which we argue is active in Malayalam and represented by the constraint WTS. The most harmonic candidate violates this constraint but due to the low weight of WTS, the violation is not fatal.

(28) *Non-Finality in co-compounds:*

	[$_{\&}$ [$_{N}$ kawi] (μ) [$_{N}$ kawajit̪ri]]	NON-FIN(FT´) 50	MATCH(ω, N) 10	*FLOAT 7	WTS 3	$\mathcal{H}$
☞	a. ($_{\omega}$ kàwíː) ($_{\omega}$ kàwajit̪rí)				1	-3
	b. ($_{\omega}$ kawìː) ($_{\omega}$ kàwajit̪rí)	1				-50
	c. ($_{\omega}$ kawìːkawajit̪rí)		1			-10
	d. ($_{\omega}$ kàwí) (μ) ($_{\omega}$ kàwajit̪rí)			1		-7

Summing up, this first case of phonological opacity in Malayalam co-compounds does not provide evidence for a stratal distinction between sub- and co-compounds. The constraint NON-FINALITY(FT´) and the prosodic parse of co-compounds conspire to compute a surface form that seems to be a case of intrastratal opacity. However, the computation remains fully parallel.

3.2 Case 2: Vowel sandhi and stress assignment

The second case of phonological opacity concerns the interaction of vowel sandhi and stress assignment. Mohanan observes that if two vowels become adjacent at a compound boundary and the nouns in the compound have Sanskrit origin, multiple vowel sandhi processes occur to resolve the vowel hiatus. Table I shows all possible vowel combinations and their surface realisation.

In almost all cases, a long vowel results, usually from fusion of the two adjacent vowels. It is this resulting long vowel, which may interact with stress assignment in Malayalam compounds. Similar to the interaction between vowel final lengthening and stress assignment, the surface long vowel that results from vowel sandhi in subcompounds attracts primary stress if it is in the second syllable of the compound. This is

V1	V2	surface
a	i	ee
a	u	oo
a	a	aa
i	i	ii
aa	au	au
i	a(i/u)	ja(i/u)
u	a(i)	ʋa(i)

Table I: Summary of vowel sandhi in Malayalam

exemplified by the subcompounding of *water*[12] and the vowel-initial noun *desire* in (29).

(29) [jaˈl èː ccʰá]

jala(m)-iccʰa

water-desire

'thirst' (subcomp) (p.120)

Analogous to the interaction of vowel final lengthening and stress assignment, the surface realisation in (29) is derived in a rule-based framework if the rules of vowel sandhi apply before the rules of stress assignment. This ensures a shifting interaction where vowel sandhi can influence the result of stress assignment.

In co-compounds, no shifting interaction is found. Instead, the fused vowel surfaces with a falling contour, see (30). This follows if stress assignment applies before vowel sandhi in co-compounds. By applying stress assignment first, the final syllable of the proper noun *Tara* receives the boundary H tone and the first syllable of the proper noun *Indran* receives the primary stress realised as L pitch. Fusion of the vowels leads to fusion of the tones and, therefore, a falling pitch contour. If the rules were in the opposite order, as they are for subcompounds, then the resulting surface form would be [t̪àːr éː n̪dránmaːrə] where secondary stress surfaces on the fused long vowel. Evidently, this is not the surface form in co-compounds.

(30) [ˈt̪àːˈr $\boxed{\text{êː}}$ n̪d̪ránmaːrə]

t̪aːra-in̪d̪ran-maːrə

Tara-Indran-PL

'Tara and Indran' (co-comp) (p.121)

As was the case in the previous section, analysing phonological opacity in co-compounds in a rule-based framework necessitates different rule orderings for sub- and co-compounds, which suggests that co-compounds undergo a different stratum of phonology compared to subcompounds. However, with the power of prosodic structure and a minimised stratal architecture, the patterns in (29) and (30) can be derived while still allowing sub- and co-compounds to be built at the same morphological stratum, the stem stratum. In a nutshell, what is crucial to our analysis is that nouns in co-compounds are parsed as separate prosodic words. This parsing, we argue, blocks vowel sandhi at the stem stratum. Then, at the word stratum, the entire co-compound plus any added inflectional affixes are parsed as a recursive prosodic word, which provides the correct environment for vowel sandhi. It follows that vowel sandhi in co-compounds is only facilitated at the word stratum after stress has been assigned at the stem stratum.

To derive the different effects of vowel sandhi in sub- and co-compounds, we introduce the high-weighted constraint in (31) below. This constraint is penalised by adjacent vowels in the same prosodic word.

(31) $*(_\omega$ V.V): Assign a violation to every sequence of featurally non-identical vowels in the same prosodic word. weight = 50

At the phonology of the stem stratum, a vowel hiatus in a subcompound structure is resolved due to the parsing of the subcompound as a single prosodic word and the high weight of $*(_\omega$ V.V), see the sketched analysis in (32) below.

(32) *Derivation of vowel sandhi and stress shifting in subcompounds:*

Stem-level: Subcompounding [$_N$ [$_N$ jala(m)] [$_N$ iccʰa]]

Stem-level Phonology: ($_\omega$ jaˈl$\boxed{\text{èː}}$ccʰá)

A vowel hiatus at the co-compound boundary is not in the same prosodic word, due to the parsing of each noun into independent prosodic words. Thus, the constraint *($_\omega$ V.V) is vacuously satisfied by co-compounds at the stem-level. Therefore, at the stem level the hiatus remains unresolved, yet the domains for stress, the minimal prosodic words, are already established which means that primary stress instantiated as a L tone, as well as the boundary H tone are associated to the relevant prosodic positions. At the word level, inflectional affixes enter the morphological computation. These affixes are prosodified in word stratum phonology, which results in the entire compound word being parsed into a recursive prosodic word. Because there is no change to the already established minimal prosodic words, the associated tones remain in their established position. However, the constraint *($_\omega$ V.V) becomes active and vowel sandhi applies, which fuses the vowels. Consequently, the H boundary tone on the final syllable of the first noun and the L tone of the primary stress on the first syllable of the second noun also fuse into a falling contour. This cyclic derivation is sketched in (33).

(33) *Cyclic derivation of vowel sandhi and stress countershifting in co-compounds:*

Stem-level: Co-compounding [$_\&$ [$_N$ t̪aːra] [$_N$ in̪d̪ran]]

Stem-level Phonology: ($_\omega$ ˈt̪àːrá) ($_\omega$ ˈìn̪d̪rán)

Word-level: Inflectional affixation [[$_\&$ ($_\omega$ ˈt̪àːrá) ($_\omega$ ˈìn̪d̪rán)] -maːr]

Word-level Phonology: ($_\omega$ ($_\omega$ ˈt̪àːré) ($_\omega$ ˈèn̪d̪rán) maːrə)

[ˈt̪àːrˈêːn̪d̪ránmaːrə]

This second case of phonological opacity thus constitutes interstratal opaque phonology, which cannot be handled intrastratally, as opposed to the first case of opacity. Mohanan (1986) takes this as evidence that the two compound structures must be built at distinct strata, thereby undergoing distinct phonologies. We show that the stratal distinction is not necessary. Instead, the phonological difference between the compounds is prosodic. In co-compounds, the prosodic parse postpones vowel sandhi until the next stratum, which results in the opaque surface form. In subcompounds, the prosodic parse immediately facilitates vowel sandhi in stem level phonology, which leads to the transparent shifting interaction.

4 Conclusion

In this paper, we provide a new perspective on Malayalam compound morphophonology which argues against Mohanan's (1986) stratal architecture and in particular argues that a stratal looping mechanism is avoidable. The central insight we contribute is that the morphological structure of subcompounds and co-compounds force different prosodic parses. The difference in prosodic parsing reflects the difference in phonological surface forms. We demonstrate that stress domains in simplex and complex compounds follow from the different prosodic parses of sub- and co-compounds. Additionally, we show that opaque phonology in co-compounds follows from two assumptions. First, a minimal stratal architecture consists of a stem-, a word- and a postlexical stratum, where all compounding takes place at the stem stratum. By virtue of this assumption, interstratal opacity is evoked by additional prosodification at the word stratum. Second, the inclusion of prosodic structure which matches morphological structure demonstrates that intrastratal opacity at the stem stratum is merely a quirk of the prosodic parsing of co-compounds and a NON-FINALITY constraint. Thus, we have shown that overwhelming empirical evidence pointing towards the need to abandon level ordering by introducing a loop is explained by prosodic structure computed by MATCH constraints in Harmonic Grammar and a minimised, restrictive stratal architecture that abides by strict level ordering.

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Notes

[1]In a stratal OT framework there are no phonological rules, rather the ranking of constraints may change from stratum to stratum

[2]All page numbers in the examples refer to Mohanan (1986) if not otherwise indicated.

[3] For reference, we include the phoneme inventory of Malayalam following Mohanan & Mohanan (1984) and Asher & Kumari (1997) using standard notations of the IPA.

	Bilabial	Dental	Alveolar	Palatoalveolar	Retroflex	Palatal	Velar	Glottal
stop	p, pʰ	t̪, t̪ʰ	t	c, cʰ	ʈ, ʈʰ		k, kʰ	
	b, bʰ	d̪, d̪ʰ		ɟ, ɟʰ	ɖ, ɖʰ		g, gʰ	
nasal	m	n̪	n	ɲ	ɳ		ŋ	
fricative	(f)		s		ʂ			h
lateral			l		ɭ			
rhotic			ɾ,r		ɻ			
glide	ʋ					j		

The segment [ɻ] is often classified independent of rhotics, as a retroflex approximant or a fricationless continuant (Mohanan 1986). We follow Sadanandan (1999) in treating it as a rhotic, as it patterns phonologically with other rhotics.Mohanan & Mohanan (1984) include palatalised velars in the phoneme inventory, Asher & Kumari (1997) indicate that almost all instances of palatalised velars are derived. The fricative [f] is found only in English loanwords. None of the fricatives in Malayalam have Dravidian origin, they are either introduced from English loanwords, like [f], or from Sanskrit. The rhotic [ɾ] is sometimes realised as a tap or a trill, a voiceless or voiced stop, when geminated it is realised as a stop [tt] (Asher & Kumari 1997). Mohanan & Mohanan (1984) derive the six surface nasals from three underlying nasals [m, n, ɲ].

	front	central	back
high	i, iː		u, uː
mid	e, eː	ə	o, oː
low		a, aː	

The mid central vowel [ə] is shown to be entirely phonologically predictable (Mohanan 1986, Mohanan 1989). It is realised as schwa or in some dialects as [ɨ] or [ɯ] (Sadanandan 1999).

[4]Mohanan (1986) notes that stress in Malayalam does not manifest itself phonetically as a change in amplitude, duration or fundamental frequency. However, vowels that are unstressed, meaning vowels that are not pronounced at L pitch, are reduced or deleted entirely. Vowels that are pronounced at L pitch can never be reduced or deleted.

[5]endeː *what* is the only bi-syllabic word that ends in a long vowel that we were able to find in Mohanan (1982, 1986). All other bi-syllabic words do not end in a long vowel, which means that final stress is not expected to surface on this final vowel given the general principles of stress assignment in Malayalam.

[6]Note that this constraint will be violated by monosyllabic words which necessarily do surface with primary stress on their final and only syllable. The realisation of primary stress on the final syllable of a monosyllabic prosodic word can be accomplished by ranking PARSE(FT)≫NON-FINALITY(FT´).

[7]Although inflectional affixes do not influence stress placement, they crucially still participate in phonological alternations like nasal assimilation and place assimilation, as well as triggering gemination in the case of, for example, the causative and transitive morphemes. Therefore, their placement within a prosodic word remains motivated.

[8]Output (17b) also violates high-weighted MATCH(Nᵐᵃˣ,ω) twice, which immediately rules this candidate out.

[9]Stratum 3 phonology according to Mohanan (1986) includes compound vowel lengthening, nasal deletion and schwa epenthesis among other phonological processes. Nasal deletion can be seen in the example (19), where the final nasal /m/ of /mat̪am/ does not surface.

[10]If the second noun in a compound has Dravidian origin then a process of initial gemination applies. This was presented in example (2) in section 1, peʈʈi- pat̪t̪aːjam -kaɭ → pèʈʈi$\boxed{\text{pp}}$at̪t̪áːjáŋŋalə. This process is restricted to subcompounds; it never occurs in co-compounds. This follows under our analysis using prosodic domains, if there is a positional restriction on gemination which disallows a geminate to surface at the left edge of a prosodic word. For a concrete implementation of compound gemination and its restriction, we refer to Author & Author (2025)

[11]The floating mora can be analysed as a linking element, similar to the way gemination is also treated as a linking element in Author & Author (2025)

[12]The noun *water* is not vowel-final underlyingly. It has an underlying final nasal, which deletes if in a compound structure. Nasal deletion in compounds feeds vowel sandhi, as the example (29) indicates.